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August 2025

AI-Driven Improvements in Patient-Doctor Communication for Remote Consultations

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Recommended Citation

Koohikamali, Mehrdad; Luiten, Lisa; Hasan, Md Mahmoudul; Kao, Chenwei; Yang, Yuan Jr; Chou, Yuhsi; Aji, Rishabh; and Roldan, Rose, "AI-Driven Improvements in Patient-Doctor Communication for Remote Consultations" (2025). *AMCIS 2025 Proceedings*. 18.
https://aisel.aisnet.org/amcis2025/health_it/sig_health/18

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Presenter Information

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AI-Driven Improvements in Patient-Doctor Communication for Remote Consultations

Emergent Research Forum (ERF) Paper

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Abstract

The fast development of telemedicine has shown the value of medical chatbots in facilitating interaction between patients and doctors. This study examines ChatDoctor, a health chatbot optimized with LLaMA, to determine the medical diagnosis suggested by Artificial intelligence compared to a physician's diagnosis. We employ 100,000+ user reviews and health chatbot conversations, applying Natural Language Processing (NLP) and Large Language Models (LLMs), to investigate patient interactions. Findings indicate that although chatbots improve access, patient engagement, and convenience, they also have accuracy and trust issues. Topic modeling identifies a contradiction in user experience, where positive comments highlight chatbot support and negative comments refer to response reliability problems. This research contributes to the Information Systems literature by demonstrating how LLMs enhance telehealth and provide valuable insights for improving chatbot design and patient-centered AI solutions. The results emphasize the need for sophisticated AI models that balance efficiency, clinical accuracy, and user demands.

Keywords

Healthcare, chatbot, telemedicine, LLM, AI.

Introduction

The COVID-19 pandemic has boosted the adoption of telemedicine to a high level across the globe. According to some studies, the telehealth usage stabilized at 38 times higher compared to the pre-pandemic period (Bestsennyy et al., 2021). The wide-ranging impacts of telemedicine and remote patient-physician interactions include more access to health care services, enhanced convenience, patient satisfaction, and overall cost savings for the patient and healthcare system. In addition, reduced healthcare worker burnout and care continuity during disasters and public health crises boost the efficiency of remote healthcare. Yet, the difficulty of medical terminology, digital divide, and issues with accurate communication are massive challenges that must be solved alongside security and privacy concerns. Diagnosis errors and mistakes in healthcare could be fatal, which implies the importance of improving communication platforms. Chatbots

and automated systems have been used in various industries to support users. In the medical field, they are finding their place to become an actual aid for medical personnel.

Artificial intelligence (AI) tools are the main foundation of chatbots. The global investment in AI for healthcare is projected to reach \$45.2 billion by 2026, up from \$4.9 billion in 2020, indicating a significant trend toward AI integration in various healthcare aspects, including communication systems and chatbots (Talby, 2021). Studies found that most patients appreciated the convenience of telemedicine, and some reported communication barriers during virtual visits (Andreadis et al., 2023). Chatbots are now included in telemedicine, and they offer various advantages that enhance patient care and simplify healthcare services, including 24/7 support, symptom screening, and triage, answering regular questions, and language the absence of physical examinations in telemedicine and chatbot interactions sometimes leads to late diagnosis or misdiagnosis, which impacts the patients' outcomes.

Large language models (LLMs) driven chatbots significantly advance telemedicine with efficient, cost-effective, and targeted patient care. They complement the abilities of healthcare professionals, improve patient engagement, and assist in more seamless healthcare delivery. However, further empirical research must be conducted on their ability to promote patient-physician communication during e-visits. Studies are needed to evaluate how AI models built on LLMs detect and identify medical conditions and provide correct diagnoses. The main objective of this study is to investigate diagnoses provided by medical chatbots and compare them with diagnoses identified by doctors. This study focuses on ChatDoctor, a medical chatbot fine-tuned to the LLaMA large language model. It starts by assessing the existing challenges in telehealth communication and identifying the role of AI. Then, this study leverages various natural language processing (NLP) and LLM methods for comprehensive data extraction centered around patient care. Finally, it investigates the AI-generated medical responses in a medical chatbot. We propose the following research questions:

1. What are the key communication barriers in telehealth settings, and to what extent do AI chatbots solve these problems in remote physician-patient communication?
2. How are advanced NLP methods—such as large language models and Named Entity Recognition—used to convert unstructured clinical information into actionable insights that inform patient-centered care?

Literature Review

Telehealth and Remote Communication

Telehealth has gained significant traction, reshaping patient-provider interactions and addressing barriers to healthcare access, especially in rural and remote areas (Shachar et al., 2020). Telehealth adoption was accelerated during the Covid-19 pandemic, highlighting regulatory and privacy challenges that require attention to ensure telehealth's sustainability. Telehealth's role in enhancing patient-doctor relationships in remote communities shows geographic isolation often limits access to primary care (Toh et al., 2016). Findings suggest that telehealth reduces these barriers, enabling patients to maintain continuous care without the need for long-distance travel, which is particularly beneficial for chronic care management (Toh et al., 2016). Successful telehealth in remote rural regions depends on both patient and provider adaptability to virtual environments and infrastructure investment in telecommunication services to support high-quality video and audio connections (Toh et al., 2016). Patient satisfaction is a crucial indicator of telehealth success, and most patients report high satisfaction levels due to the convenience and time-saving benefits of virtual visits (Kruse et al., 2017). However, satisfaction can vary depending on the telehealth system's reliability and user-friendliness, and the perceived quality of provider communication (Kruse et al., 2017). Patients' experiences with clinical video telehealth visits show communication challenges, particularly for patients who feel distanced from the clinical conversation (Gordon et al., 2020). Telehealth offers convenience, but it also requires deliberate communication strategies to make patients feel engaged (Gordon et al., 2020).

AI Chatbots in Healthcare

AI chatbots are revolutionizing customer service, healthcare, mental health, and e-commerce by enhancing engagement, accessibility, and efficiency. These chatbots facilitate continuous user interaction by providing

24/7 support and swift responses, which are instrumental in improving customer satisfaction and engagement (Casheekar et al., 2024). A chatbot is an application that uses artificial intelligence techniques to chat with the user in a way similar to human conversation. (Hill et al., 2015). Chatbots typically accept input messages from the user—typing, spoken words, or menu selects—and provide suitable responses based on their training set, language patterns, or predefined rules. Modern-day chatbots often leverage machine learning and NLP/LLM capabilities to interpret context, identify intent, and adapt through experience over time, allowing them to support more advanced questions and offer tailored experiences. Chatbots offer essential support for patient management, chronic disease care, and mental health services in healthcare. Medical chatbots provide valuable patient support, offering resources such as reminders, educational materials, and behavioral interventions for chronic disease management. (Laymouna et al., 2024). Mental health-focused chatbots have integrated AI-driven cognitive behavioral therapy (CBT) techniques, helping users manage anxiety, depression, and other mental health challenges (de Filippis & Al Foyssal, 2024). These chatbots are often equipped with mood tracking and mindfulness exercises that provide psychoeducation and guide users in self-care strategies when human support is inaccessible or unaffordable (Denecke et al., 2021). Chatbots have shown potential to bridge gaps in mental health care, offering immediate support to individuals and freeing healthcare providers to focus on critical cases.

Healthcare companies amass tons of data every day, ranging from prescriptions to medical history, and more (Clark & Bailey, 2024). Big medical data includes a combination of unstructured and coded data such as medical imaging data, experimental data, and medical text data (Xiao et al., 2021). Utilizing the textual content with the structured data can improve healthcare services in many ways, such as identifying new relationships between described symptoms and documented medical history with new diagnoses (Ramon-Gonen et al., 2023). For physicians, incorporating a text-based analysis can help them to achieve a more nuanced and accurate understanding of diagnoses, instead of relying solely on the strictly structured International Classification of Diseases – 10th version (ICD-10) coding format (Ford et al., 2016). Most current research tries to identify whether the textual data decreases medical errors, reduces costs, or improves healthcare quality. In data analytics, the bulk of studies focus on using NLP and LLMs to analyze and interpret healthcare data.

Research Model and Methodology

Media synchronicity theory (MST) provides a helpful framework for better understanding communication in telehealth, especially when using technologies like chatbots. MST says that communication works best when the communication tool matches the needs of the task (Dennis et al., 2008). It identifies two key processes: conveyance (sharing new information) and convergence (building shared understanding). According to MST, low-synchronicity tools (like emails or text chats) are better for conveyance, while high-synchronicity tools (like video calls) are better for convergence (Dennis et al., 2008). The MST framework is applicable for this study since it specifically positions chatbots within the MST framework, considering their typical characteristics. MST can both the potential benefits (conveyance support) and limitations (convergence challenges) of chatbots in patient-doctor communication.

The methodology has four main steps. The first step is understanding telehealth barriers and AI chatbots, which will require peer-reviewed papers and online user reviews. We used “telehealth” and “communication” as two main keywords to find articles that are published in top Information Systems and National Institutes of Health (NIH) journals since 2016. Second, there is large-scale data processing and summarization, which is based on existing online chat medical conversations. Third, there is detailed information structuring and analysis, which requires the same medical chat conversations. Finally, the evaluation and strategic insights will require the results of the developed methods. We will use manually coded information for model evaluation.

For step one, we gathered 100 peer-reviewed articles to investigate the communication patterns in telehealth. The articles' publication dates range from 2017 to 2024. We then collected the top seven medical applications on Google Play and retrieved the user reviews, which total more than 100,000. For steps two and three, several conversational medical interactions are used. The primary source is from a team that created "Chat Doctor," which gathered anonymized conversations from two telemedicine companies, and another dataset from health-oriented AI chatbots. These telemedicine websites, Healthcare Magic (India) and iCliniq (United States), provide patients access to certified doctors for live consultation, focusing on

access through websites and mobile apps. Both sites employ subscription models, have private funding, and are independent of large pharmaceutical firms. Previously, the Chat Doctor team pre-cleaned, grammar-checked, and de-identified the data they applied to train their medical chatbot. The iCliniq dataset presents real patient-doctor interactions with GPT-based model answers (GenMed GPT) and Chat Doctor. GenMed GPT is a health application, combined with a disease database, to present more comprehensive medical information during AI-based consultations. The data consists of medical chatbot conversations collected in 2023. The sample consists of 100,000 conversations from Healthcare Magic, 10,000 conversations from iCliniq, and 5,000 generated AI conversations from GenMedGPT and the disease database. The following table provides an overview of the combined sample for Transcript data (patient-LLM interactions) and Symptom data (disease-symptom relationship).

Dataset	Column	Meaning/Description	Example
Transcript	Instruction	Prompt provided to GPT or LLM to guide the response.	If you are a doctor, please answer the medical questions based on the patient's description.
Transcript	Input	Patient's concern or symptoms as stated.	Hi Doctor. I think I might have Turner syndrome.
Transcript	Output/Answer	Advice or consultation text generated by the LLM.	We will need to do a complete physical exam, ultrasonography, echocardiography, etc.
Symptom	Disease	Name of the disease.	Endophthalmitis
Symptom	Symptoms	List of external signs or manifestations associated with the disease.	['Pain in eye', 'Eye redness', 'Swollen eye', 'Diminished vision', 'Itchiness of eye', ...]

Table 1. Transcript and Symptom Data Sample

This study begins by investigating telehealth communication patterns and obstacles to determine the root of problems in virtual health settings. It then progresses to investigating how AI chatbots assist in correcting these, unveiling the advantages and disadvantages of automated interaction.

Preliminary Analysis and Results

Analyzing the collected publications provides insights about the top barriers to telemedicine, including data privacy, ethical concerns, Misinformation, data accuracy, implementation costs, user trust, and a one-size-fits-all approach. Further analysis will provide a knowledge graph of existing research. Topic modelling analysis of reviews of mobile applications provides information about the common themes from the user perspective. We used Latent Dirichlet allocation (LDA), and the themes derived from four and five-star feedback ratings emphasize the assistance, utility, and emotional support provided by the app. Conversely, the themes derived for low rating feedback are the opposite of those derived from positive feedback, indicating differences in user experience based on expectation and accomplishment. Users find the chatbot to be helpful, with many citing its ability to assist them ("help," "helpful," "helped") and be informative ("informative," "information"). Many also mention its therapeutic value, describing how it makes them feel (better, "feel," "amazing," "love"). The chatbot is also appreciated for being convenient and easy to use ("use," "experience"). However, there are challenges with the chatbot's responses, with some doubting its efficacy and reliability ("talk," "work," "information"). Others simply find the assistance it provides not meaningful or effective enough ("feel," "pretty," "far"). There are also technical limitations and usability issues in that some users find the application hard to use ("application," "use"). We have coded stress levels for the Reddit data sample (200 records) to use for the model evaluation. Researchers developed a coding scheme based on established psychological stress indicators. Two authors with a Psychology background helped code samples. The high inter-reliability validates the reliability of the method.

Conclusion and Expected Contributions

This study will offer several contributions to the area of healthcare analytics and LLM. This study will show the key role of analyzing patients' descriptions of their condition and appropriate actions, including AI diagnosis. Preliminary findings suggest that while AI chatbots enrich telemedicine through 24/7 consultation, symptom assessment, and triage, they also face accuracy, personalization, and trust concerns

in patient communication. User review topic modeling analysis reflects the benefits and disadvantages of such AI technologies that denote the division in user experience in relation to expectation and outcome. While chatbots have proved useful, informative, and supportive for most users, there are some users concerned with response correctness, technical issues, and engagement quality. This study will help to shed some light on how medical chatbots can incorporate AI and use expert opinions in critical and high-risk conditions. We employ a combination of NLP and LLM methodologies to investigate AI-generated comments and compare them with physicians' responses. Theoretically, this study benefits Information Systems research by demonstrating how LLMs and NLP techniques enhance telehealth conversations. It bridges the gap between AI chatbots and patient-provider communication and offers a contribution to Expectation-Confirmation Theory in the adoption of healthcare AI. In practice, the findings offer actionable guidance for developers and healthcare providers to improve AI chatbot precision, user experience, and regulatory compliance, ensuring that AI-driven telemedicine solutions provide more personalized, efficient, and reliable patient care.

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